# Project 02 — Multi-Site Expansion + DHCP

Branch Site, WAN Link, Centralized DHCP, IPv6 Dual-Stack, DNS

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Platform: Cisco CML 2.9 | IOL + IOL-L2

Prerequisite: Project 01 — Campus Foundation

Methodology: Build → Verify → Break → Fix

# Setup and Scenario

## Business Context

The HQ campus from Project 01 exists as an isolated single site: static IPs on all endpoints, no DHCP or DNS services, and no branch connectivity. A new branch office needs to be connected, all endpoints should get IPs automatically, and IPv6 dual-stack should be enabled for forward compatibility.

## What You Will Build

- Branch site switching infrastructure (BR-RTR1, BR-DSW1, BR-ASW1) matching HQ's VLAN scheme
- Point-to-point /30 WAN link between HQ-RTR1 and BR-RTR1
- Static routing so both sites reach all remote subnets including management
- Centralized Dnsmasq DHCP server at HQ (10.1.99.50) serving 8 VLAN pools via ip helper-address relay
- Static DHCP reservations by MAC for branch endpoints
- IPv6 dual-stack: /126 WAN, /64 VLAN 100 at both sites, SLAAC for endpoints
- DNS A records for all devices in lab.local domain

# STAR Summary

**Situation:** HQ campus existed as an isolated single site with static IPs on all endpoints and no DHCP or DNS service.

**Task:** Add a branch site, connect both sites over WAN, deploy centralized DHCP serving all 8 VLANs via relay, add IPv6 dual-stack, and verify DNS end-to-end.

**Action:** Deployed BR-RTR1/DSW1/ASW1 matching HQ's VLAN scheme, built /30 WAN, deployed Dnsmasq DHCP server at HQ serving 8 pools via ip helper-address, configured IPv6 SLAAC, added DNS A records for all devices.

**Result:** Can design a multi-site enterprise network, configure centralized DHCP serving remote sites over WAN, and verify dual-stack connectivity from endpoints at both sites.

# Technologies Used

- Router-on-a-Stick (branch site)
- 802.1Q Trunking
- Point-to-Point WAN /30 link
- Static Routing (IPv4)
- Centralized DHCP (Dnsmasq) with ip helper-address relay
- DHCP pools for 8 VLANs across both sites
- Static DHCP reservations by MAC address

- IPv6 dual-stack (/126 WAN / /64 VLAN 100)
- SLAAC via Router Advertisements
- DNS with address= directives (Dnsmasq)
- Rapid-PVST+ STP, PortFast + BPDU Guard
- SSH v2 hardening (all devices)
- ip routing on access switch (BR-ASW1)

## Network Topology

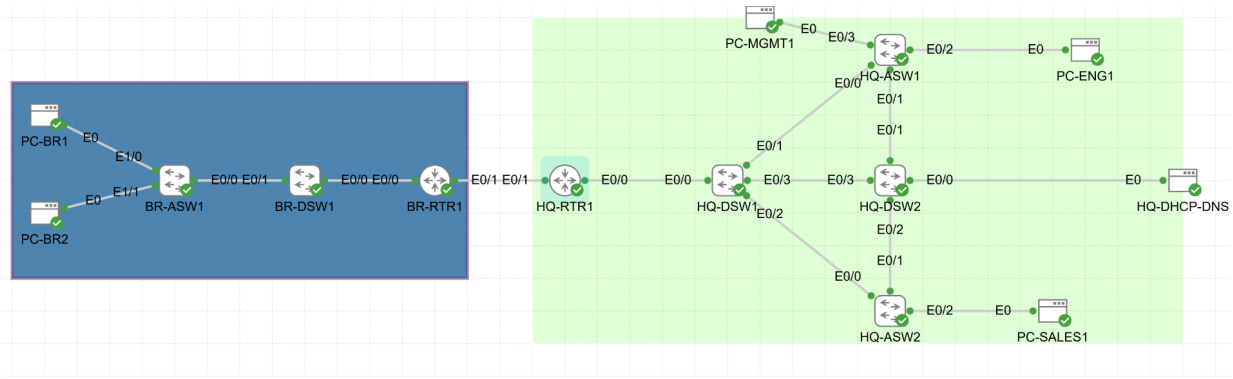


*Figure T.1 — CML Topology — Project 02 Multi-Site Expansion*

## IP Addressing

### Branch Site (10.2.x.x)

| Device | Interface | IP Address | Subnet | Purpose |
|---|---|---|---|---|
| BR-RTR1 | E0/0.100 | 10.2.100.1 | /24 | Engineering gateway |
| BR-RTR1 | E0/0.200 | 10.2.200.1 | /24 | Sales gateway |
| BR-RTR1 | E0/0.300 | 10.2.44.1 | /24 | Guest gateway |
| BR-RTR1 | E0/0.500 | 10.2.50.1 | /24 | Voice gateway |
| BR-RTR1 | E0/0.999 | 10.2.99.1 | /24 | Management gateway |
| BR-DSW1 | VLAN 999 | 10.2.99.2 | /24 | Management SVI |
| BR-ASW1 | VLAN 999 | 10.2.99.3 | /24 | Management SVI |
| PC-BR1 | eth0 | 10.2.100.197 | /24 | Branch Engineering (DHCP reserved) |
| PC-BR2 | eth0 | 10.2.200.108 | /24 | Branch Sales (DHCP reserved) |

## WAN Link

| Device | Interface | IP Address | Subnet | Purpose |
|---|---|---|---|---|
| HQ-RTR1 | E0/1 | 10.0.0.1 | /30 | WAN HQ-side |
| BR-RTR1 | E0/1 | 10.0.0.2 | /30 | WAN Branch-side |

## HQ Additions

| Device | Interface | IP Address | Subnet | Purpose |
|---|---|---|---|---|
| HQ-RTR1 | E0/0.500 | 10.1.50.1 | /24 | Voice VLAN gateway (new) |
| HQ-DHCP-DNS | eth0 | 10.1.99.50 | /24 | Centralized DHCP+DNS |

## IPv6 Addressing

| Interface | IPv6 Address | Purpose |
|---|---|---|
| HQ-RTR1 E0/1 | 2001:db8:0:1::1/126 | WAN HQ-side |
| BR-RTR1 E0/1 | 2001:db8:0:1::2/126 | WAN Branch-side |
| HQ-RTR1 E0/0.100 | 2001:db8:1:100::1/64 | HQ Eng IPv6 gateway |
| BR-RTR1 E0/0.100 | 2001:db8:2:100::1/64 | Branch Eng IPv6 gateway |

## DHCP Pools

| Tag | Subnet | Range | Gateway | DNS |
|---|---|---|---|---|
| hq-eng | 10.1.100.0/24 | .100–.200 | 10.1.100.1 | 10.1.99.50 |
| hq-sales | 10.1.200.0/24 | .100–.200 | 10.1.200.1 | 10.1.99.50 |
| hq-guest | 10.1.44.0/24 | .100–.200 | 10.1.44.1 | 10.1.99.50 |
| hq-voice | 10.1.50.0/24 | .100–.200 | 10.1.50.1 | 10.1.99.50 |
| br-eng | 10.2.100.0/24 | .100–.200 | 10.2.100.1 | 10.1.99.50 |
| br-sales | 10.2.200.0/24 | .100–.200 | 10.2.200.1 | 10.1.99.50 |
| br-guest | 10.2.44.0/24 | .100–.200 | 10.2.44.1 | 10.1.99.50 |
| br-voice | 10.2.50.0/24 | .100–.200 | 10.2.50.1 | 10.1.99.50 |

## Phase 1: Branch Site Base

**Goal:** Build the complete branch switching and routing infrastructure from scratch.

### Step 1

On BR-RTR1, configure router-on-a-stick for all branch VLANs:

```
enable
configure terminal
hostname BR-RTR1
lldp run
interface Ethernet0/0
 description TRUNK-TO-BR-DSW1-E0/0
 no ip address
 no shutdown
interface Ethernet0/0.100
 description GATEWAY-VLAN100-ENGINEERING
 encapsulation dot1Q 100
 ip address 10.2.100.1 255.255.255.0
 no shutdown
interface Ethernet0/0.200
 description GATEWAY-VLAN200-SALES
 encapsulation dot1Q 200
 ip address 10.2.200.1 255.255.255.0
 no shutdown
interface Ethernet0/0.300
 description GATEWAY-VLAN300-GUEST
 encapsulation dot1Q 300
 ip address 10.2.44.1 255.255.255.0
 no shutdown
interface Ethernet0/0.500
 description GATEWAY-VLAN500-VOICE
 encapsulation dot1Q 500
 ip address 10.2.50.1 255.255.255.0
 no shutdown
interface Ethernet0/0.999
 description GATEWAY-VLAN999-MGMT
 encapsulation dot1Q 999
 ip address 10.2.99.1 255.255.255.0
 no shutdown
interface Ethernet0/0.1000
 description NATIVE-VLAN1000-UNUSED
 encapsulation dot1Q 1000 native
 no shutdown
```

```
end
write memory
```

> **WHY:** *Voice VLAN 500 is built now even without phones — avoids rework when Project 11 (QoS) is configured. The native VLAN 1000 subinterface explicitly tags untagged frames, preventing VLAN hopping attacks.*

### Step 2

On BR-DSW1, configure as distribution switch and STP root:

```
enable
configure terminal
hostname BR-DSW1
lldp run
ip routing
vlan 100
 name Engineering
vlan 200
 name Sales
vlan 300
 name Guest
vlan 500
 name Voice
vlan 999
 name Management
vlan 1000
 name NATIVE-UNUSED
interface vlan 999
 ip address 10.2.99.2 255.255.255.0
 no shutdown
ip default-gateway 10.2.99.1
spanning-tree mode rapid-pvst
spanning-tree vlan 100,200,300,500,999,1000 priority 4096
interface Ethernet0/0
 description TRUNK-TO-BR-RTR1-E0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 1000
 switchport trunk allowed vlan 100,200,300,500,999,1000
 no shutdown
interface Ethernet0/1
 description TRUNK-TO-BR-ASW1-E0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 1000
```

```
 switchport trunk allowed vlan 100,200,300,500,999,1000
 no shutdown
end
write memory
```

**WHY:** *ip routing enabled: Project 03 OSPF needs BR-DSW1 as an area router — enabling it now avoids reconfiguration. Single switch as root for all VLANs: Only one distribution switch at branch — auto-election could make BR-ASW1 root, causing a forwarding black-hole.*

### Step 3

On BR-ASW1, configure access switch with ip routing:

```
enable
configure terminal
hostname BR-ASW1
lldp run
ip routing
ip route 0.0.0.0 0.0.0.0 10.2.99.1
vlan 100
 name Engineering
vlan 200
 name Sales
vlan 300
 name Guest
vlan 500
 name Voice
vlan 999
 name Management
vlan 1000
 name NATIVE-UNUSED
interface vlan 999
 ip address 10.2.99.3 255.255.255.0
 no shutdown
spanning-tree mode rapid-pvst
interface Ethernet0/0
 description TRUNK-TO-BR-DSW1-E0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 1000
 switchport trunk allowed vlan 100,200,300,500,999,1000
 no shutdown
interface Ethernet1/0
 description ACCESS-PC-BR1-VLAN100
 switchport mode access
 switchport access vlan 100
```

```
 switchport nonegotiate
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown
interface Ethernet1/1
 description ACCESS-PC-BR2-VLAN200
 switchport mode access
 switchport access vlan 200
 switchport nonegotiate
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown
end
write memory
```

**WHY:** *ip routing instead of ip default-gateway: ip default-gateway cannot route replies destined to remote subnets. When HQ-RTR1 (10.0.0.1) pings BR-ASW1, the ICMP reply must return to a remote subnet — this requires the routing engine, not the management-plane handler.*

**NOTE:** BR-ASW1 uses BOTH Ethernet0/x (trunks) AND Ethernet1/x (access ports). Always verify with show cdp neighbors after cabling.

## Verification

```
show ip interface brief          ! BR-RTR1 — E0/0.100 through .999 all up/up
show interfaces trunk            ! BR-DSW1 — both trunks up, native 1000
show spanning-tree vlan 100      ! BR-DSW1 — "This bridge is the root"
```

*[Image not found: /home/leonel/code/enterprise-network-labs/project-01-campus-foundation/verification/screenshots/p02-p1-02-RTR1-ip-brief.png]*

*Figure 1.1 — BR-RTR1 show ip interface brief*

## Phase 2: WAN Connectivity

**Goal:** Connect HQ and Branch over a point-to-point /30 link and add static routes so both sites reach all remote subnets.

### Step 1

On HQ-RTR1, configure the WAN interface and add VLAN 500 (Voice) subinterface:

```
configure terminal
interface Ethernet0/1
 description WAN-TO-BR-RTR1-E0/1
 ip address 10.0.0.1 255.255.255.252
 bandwidth 1000
 delay 1000
 no shutdown
interface Ethernet0/0.500
 description GATEWAY-VLAN500-VOICE
 encapsulation dot1Q 500
 ip address 10.1.50.1 255.255.255.0
ip route 10.2.100.0 255.255.255.0 10.0.0.2
ip route 10.2.200.0 255.255.255.0 10.0.0.2
ip route 10.2.44.0  255.255.255.0 10.0.0.2
ip route 10.2.50.0  255.255.255.0 10.0.0.2
ip route 10.2.99.0  255.255.255.0 10.0.0.2
end
write memory
```

> **WHY:** */30 gives exactly 2 usable IPs for a point-to-point link — using /24 wastes 252 addresses. bandwidth 1000 and delay 1000: OSPF in Project 03 calculates cost from bandwidth — IOL defaults to 10 Mbps making WAN look identical to LAN. Setting 1 Mbps now ensures correct OSPF cost without rework. 10.2.99.0 route included: Without it, SSH to branch switches fails.*

### Step 2

On BR-RTR1, configure the WAN interface and static routes back to HQ:

```
configure terminal
interface Ethernet0/1
 description WAN-TO-HQ-RTR1-E0/1
 ip address 10.0.0.2 255.255.255.252
 bandwidth 1000
 delay 1000
 no shutdown
ip route 10.1.100.0 255.255.255.0 10.0.0.1
ip route 10.1.200.0 255.255.255.0 10.0.0.1
```

```
ip route 10.1.44.0  255.255.255.0 10.0.0.1
ip route 10.1.50.0  255.255.255.0 10.0.0.1
ip route 10.1.99.0  255.255.255.0 10.0.0.1
end
write memory
```

### Verification

```
show ip route            ! HQ-RTR1 — 5 static routes to 10.2.x.x via 10.0.0.2
ping 10.2.99.3           ! HQ-RTR1 — cross-site ping to BR-ASW1 management
```

*[Image not found: /home/leonel/code/enterprise-network-labs/project-01-campus-foundation/verification/screenshots/p02-p2-01-cross-site-ping.png]*

*Figure 2.1 — Cross-site ping confirmed*

## Phase 3: Centralized DHCP

**Goal:** Deploy HQ-DHCP-DNS (Dnsmasq) at 10.1.99.50 and configure ip helper-address relay on both routers to serve all 8 VLANs across both sites.

### Step 1

In CML, cable HQ-DHCP-DNS eth0 to HQ-DSW2 Ethernet0/0. Then on HQ-DSW2, configure the port as an access port in VLAN 999:

```
configure terminal
interface Ethernet0/0
 description TO_HQ-DHCP-DNS
 switchport mode access
 switchport access vlan 999
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown
end
write memory
```

> **WHY:** *The DHCP server has IP 10.1.99.50 on the management subnet — it must be in VLAN 999 to have Layer 2 connectivity.*

### Step 2

In CML, click HQ-DHCP-DNS → CONFIG tab. Paste the following as the Startup Script:

```
#!/bin/sh
hostname hq-dhcp-dns
ip addr add 10.1.99.50/24 dev eth0
ip link set eth0 up
ip route add default via 10.1.99.1
kill -9 $(cat /var/run/dnsmasq.pid) 2>/dev/null
dnsmasq --conf-file=/etc/dnsmasq.conf
echo "=== HQ-DHCP-DNS READY ==="
ip addr show eth0
ip route show
```

### Step 3

In the CML CONFIG tab, paste the following as dnsmasq.conf:

```
domain=lab.local
local=/lab.local/
log-dhcp
dhcp-range=set:hq-eng,10.1.100.100,10.1.100.200,255.255.255.0,24h
dhcp-option=tag:hq-eng,option:router,10.1.100.1
```

```
dhcp-option=tag:hq-eng,option:dns-server,10.1.99.50
dhcp-range=set:hq-sales,10.1.200.100,10.1.200.200,255.255.255.0,24h
dhcp-option=tag:hq-sales,option:router,10.1.200.1
dhcp-option=tag:hq-sales,option:dns-server,10.1.99.50
dhcp-range=set:hq-guest,10.1.44.100,10.1.44.200,255.255.255.0,24h
dhcp-option=tag:hq-guest,option:router,10.1.44.1
dhcp-option=tag:hq-guest,option:dns-server,10.1.99.50
dhcp-range=set:hq-voice,10.1.50.100,10.1.50.200,255.255.255.0,24h
dhcp-option=tag:hq-voice,option:router,10.1.50.1
dhcp-option=tag:hq-voice,option:dns-server,10.1.99.50
dhcp-range=set:br-eng,10.2.100.100,10.2.100.200,255.255.255.0,24h
dhcp-option=tag:br-eng,option:router,10.2.100.1
dhcp-option=tag:br-eng,option:dns-server,10.1.99.50
dhcp-range=set:br-sales,10.2.200.100,10.2.200.200,255.255.255.0,24h
dhcp-option=tag:br-sales,option:router,10.2.200.1
dhcp-option=tag:br-sales,option:dns-server,10.1.99.50
dhcp-range=set:br-guest,10.2.44.100,10.2.44.200,255.255.255.0,24h
dhcp-option=tag:br-guest,option:router,10.2.44.1
dhcp-option=tag:br-guest,option:dns-server,10.1.99.50
dhcp-range=set:br-voice,10.2.50.100,10.2.50.200,255.255.255.0,24h
dhcp-option=tag:br-voice,option:router,10.2.50.1
dhcp-option=tag:br-voice,option:dns-server,10.1.99.50
dhcp-host=52:54:00:22:a6:9c,pc-br1,10.2.100.197
dhcp-host=52:54:00:53:da:69,pc-br2,10.2.200.108
address=/hq-rtr1.lab.local/10.1.99.1
address=/hq-dsw1.lab.local/10.1.99.11
address=/hq-dsw2.lab.local/10.1.99.12
address=/hq-asw1.lab.local/10.1.99.13
address=/hq-asw2.lab.local/10.1.99.14
address=/br-rtr1.lab.local/10.2.99.1
address=/br-dsw1.lab.local/10.2.99.2
address=/br-asw1.lab.local/10.2.99.3
address=/hq-dhcp-dns.lab.local/10.1.99.50
address=/pc-br1.lab.local/10.2.100.197
address=/pc-br2.lab.local/10.2.200.108
```

**NOTE:** CRITICAL: Do NOT include bind-interfaces — it causes a container crash (exit code 30). Do NOT include interface=eth0 — it silently blocks relayed DHCP packets.

### Step 4

On both HQ-RTR1 and BR-RTR1, configure ip helper-address on all data VLAN subinterfaces:

```
configure terminal
interface Ethernet0/0.100
```

```
 ip helper-address 10.1.99.50
interface Ethernet0/0.200
 ip helper-address 10.1.99.50
interface Ethernet0/0.300
 ip helper-address 10.1.99.50
interface Ethernet0/0.500
 ip helper-address 10.1.99.50
end
write memory
```

> **WHY:** *The relay inserts the subinterface IP as giaddr (gateway address field). Dnsmasq reads giaddr to select the correct DHCP pool. If configured on parent E0/0, giaddr = 0.0.0.0 and pool selection fails.*

### Verification

Wait 20 seconds after HQ-DHCP-DNS shows BOOTED (startup script needs time). Then on PC-BR1:

```
udhcpc -i eth0 -f
```

Expected: IP in range 10.2.100.100–200 from server 10.1.99.50.

*[Image not found: /home/leonel/code/enterprise-network-labs/project-01-campus-foundation/verification/screenshots/p02-p3-01-dnsmasq-log.png]*

*Figure 3.1 — Dnsmasq DHCP leases across both sites*

## Phase 4: IPv6 Dual-Stack

**Goal:** Add IPv6 to the WAN link and VLAN 100 at both sites. Enable SLAAC so endpoints self-configure IPv6 addresses.

### Step 1

On HQ-RTR1, enable IPv6 routing and configure interfaces:

```
configure terminal
ipv6 unicast-routing
interface Ethernet0/1
 ipv6 address 2001:db8:0:1::1/126
 ipv6 enable
interface Ethernet0/0.100
 ipv6 address 2001:db8:1:100::1/64
 ipv6 enable
ipv6 route 2001:db8:2:100::/64 2001:db8:0:1::2
end
write memory
```

**WHY:** *ipv6 unicast-routing: Without this global command, the router drops all IPv6 packets even if interfaces have addresses. /126 on WAN: IPv6 equivalent of /30 — exactly 2 usable addresses. /64 on VLAN 100: SLAAC requires /64 — hosts use EUI-64 (derived from MAC) combined with the /64 prefix to build a global IPv6 address automatically.*

### Step 2

On BR-RTR1, mirror the IPv6 configuration:

```
configure terminal
ipv6 unicast-routing
interface Ethernet0/1
 ipv6 address 2001:db8:0:1::2/126
 ipv6 enable
interface Ethernet0/0.100
 ipv6 address 2001:db8:2:100::1/64
 ipv6 enable
ipv6 route 2001:db8:1:100::/64 2001:db8:0:1::1
end
write memory
```

### Verification

```
show ipv6 route                    ! HQ-RTR1 — static route to 2001:db8:2:100::/64
ping ipv6 2001:db8:0:1::2          ! HQ-RTR1 to BR-RTR1 WAN
```

```
ip -6 addr show eth0                ! PC-BR1 — SLAAC address in 2001:db8:2:100::/64
```

*[Image not found: /home/leonel/code/enterprise-network-labs/project-01-campus-foundation/verification/screenshots/p02-p4-03-cross-site-ipv6-ping.png]*

*Figure 4.1 — Cross-site IPv6 ping*

# Phase 5: DNS End-to-End

**Goal:** Enable DNS resolution for all devices using the Dnsmasq server.

### Step 1

Phase 1 applied no ip domain lookup to stop IOS from hanging on mistyped commands. Re-enable DNS now that a valid server exists. On HQ-RTR1 and BR-RTR1:

```
configure terminal
ip domain lookup
ip name-server 10.1.99.50
end
write memory
```

> **WHY:** *no ip domain lookup blocks ALL DNS queries from the router — including legitimate ones like ping hq-rtr1.lab.local. With a real name-server configured, this restriction no longer applies.*

### Verification

```
nslookup hq-rtr1.lab.local 10.1.99.50    ! PC-BR1 — should return 10.1.99.1
nslookup br-rtr1.lab.local 10.1.99.50    ! PC-BR1 — should return 10.2.99.1
ping hq-rtr1.lab.local                   ! BR-RTR1 — should resolve and ping
```

*[Image not found: /home/leonel/code/enterprise-network-labs/project-01-campus-foundation/verification/screenshots/p02-p5-01-PC-BR1-nslookup-hqrtr1.png]*

*Figure 5.1 — DNS resolution from PC-BR1*

## Final Verification Checklist

| # | Test | Command | Device | Expected |
|---|---|---|---|---|
| 1 | Branch subinterfaces | show ip interface brief | BR-RTR1 | E0/0.100–.999 all up/up |
| 2 | Branch trunks | show interfaces trunk | BR-DSW1 | E0/0 and E0/1 trunking, native 1000 |
| 3 | Branch STP root | show spanning-tree vlan 100 | BR-DSW1 | This bridge is the root |
| 4 | WAN routing | show ip route | HQ-RTR1 | 5 static routes to 10.2.x.x |
| 5 | Cross-site ping | ping 10.2.99.3 | HQ-RTR1 | 5/5 success |
| 6 | DHCP relay | udhcpc -i eth0 | PC-BR1 | IP from 10.1.99.50 in 10.2.100.x |
| 7 | Helper-address audit | show run \| section helper | Both RTRs | 10.1.99.50 on all data subifs |
| 8 | IPv6 WAN | ping ipv6 2001:db8:0:1::2 | HQ-RTR1 | 5/5 success |
| 9 | SLAAC address | ip -6 addr show eth0 | PC-BR1 | 2001:db8:2:100::/64 present |
| 10 | DNS resolution | nslookup hq-rtr1.lab.local | PC-BR1 | Returns 10.1.99.1 |

## Troubleshooting — Top 5 Issues

### Issue 1 — Dnsmasq Container Crash (bind-interfaces)

**Symptom:** CML reported "container failed to start" with exit code 30 when starting HQ-DHCP-DNS.

**Root Cause:** The bind-interfaces directive causes dnsmasq to crash because it tries to bind to eth0 before the interface is fully available — a race condition on CML Dnsmasq nodes.

**Fix:** Removed bind-interfaces from dnsmasq.conf and used the RESTORE button in the CML CONFIG tab to apply the corrected configuration.

### Issue 2 — ip default-gateway Insufficient for Remote-Subnet ICMP

**Symptom:** Pings from HQ-RTR1 (10.0.0.1) to BR-ASW1 (10.2.99.3) failed with 0% success while pings to BR-RTR1 and BR-DSW1 succeeded.

**Root Cause:** ip default-gateway on IOL-L2 cannot route ICMP replies back to remote subnets. It only handles management-plane packets destined to the same subnet.

**Fix:** Replaced ip default-gateway with ip routing and added ip route 0.0.0.0 0.0.0.0 10.2.99.1.

### Issue 3 — interface=eth0 Blocking DHCP Relay Packets

**Symptom:** After configuring helper-address, dnsmasq logs continued showing "no address range available via eth0" instead of "via 10.x.x.x" — indicating relay was not recognized.

**Root Cause:** interface=eth0 restricts dnsmasq to only process broadcast DHCP requests on eth0. Relayed packets arrive as unicast UDP — this directive silently dropped them.

**Fix:** Removed interface=eth0 from dnsmasq.conf.

## Issue 4 — HQ-DHCP-DNS NO-CARRIER — Missing VLAN Config on HQ-DSW2

**Symptom:** ip link show eth0 on HQ-DHCP-DNS showed NO-CARRIER, state DOWN despite a cable existing in CML.

**Root Cause:** HQ-DSW2 E0/0 (connected to HQ-DHCP-DNS) was never configured as an access port. The port was in default VLAN 1 with no active VLAN assignment.

**Fix:** Configured HQ-DSW2 E0/0 as access port VLAN 999 with PortFast and BPDU Guard.

## Issue 5 — Null0 Black-Hole Masking DHCP Failure

**Symptom:** Two simultaneous DHCP failures (Branch VLAN 100, HQ VLAN 200). Ping to DHCP server appeared to work intermittently — false positive.

**Root Cause:** Three faults — wrong helper-address (10.1.99.99) on two subinterfaces, PLUS a Null0 static route competing with the correct route to 10.1.99.0/24. IOS load-balanced across both routes; relay packets consistently hit the Null0 path.

**Fix:** Removed Null0 route FIRST (dependency order matters), then corrected helper-address to 10.1.99.50 on both subinterfaces.

## Break/Fix Challenge

**Scenario:** Two helpdesk tickets — Branch VLAN 100 no DHCP, HQ VLAN 200 no DHCP. DNS and management unaffected.

### Step 1

Verify endpoints cannot get DHCP:

```
udhcpc -i eth0 -f    ! PC-BR1 — should time out
```

### Step 2

Check Layer 1/2 (start at the bottom of OSI):

```
show vlan brief          ! both sites — VLANs active?
show interfaces trunk    ! trunks up, correct VLANs?
```

Result: Clean. No L1/L2 issues.

### Step 3

Check routing table for DHCP server subnet:

```
show ip route 10.1.99.0    ! BR-RTR1
```

Look for two competing entries — one via 10.0.0.1 (correct) and one via Null0 (black-hole). Intermittent ping success is a FALSE POSITIVE.

### Step 4

Check helper-address config:

```
show running-config | section helper    ! both routers
```

Look for 10.1.99.99 instead of 10.1.99.50 on affected subinterfaces.

### Step 5

Apply fixes in dependency order:

```
! Fix 1: Remove Null0 route FIRST
no ip route 10.1.99.0 255.255.255.0 Null0
! Fix 2: Correct helper on BR-RTR1
interface Ethernet0/0.100
 ip helper-address 10.1.99.50
! Fix 3: Correct helper on HQ-RTR1
interface Ethernet0/0.200
 ip helper-address 10.1.99.50
```

*[Image not found: /home/leonel/code/enterprise-network-labs/project-01-campus-foundation/verification/screenshots/p02-bf-05-PC-BR1-dhcp-restored.png]*

*Figure B.1 — DHCP restored after all three faults fixed*

**NOTE:** Key Lesson: Fix lowest-layer faults first. Fixing helper-address while the Null0 route exists still fails — the routing fault blocks the fix from taking effect.